

Discharge Forecaster

Model #2 of the AquaGuard / Bangladesh Flood Early-Warning System

A river-discharge regression forecaster — literature review, methodology, real data,
and evaluation results

August 14, 2026

Companion to the flood-risk-classifier and flood-susceptibility reports

1. Introduction & Purpose

Built as a pivot alongside, not instead of, the flood-risk classifier (model #1): rather than a binary flood/no-flood flag, this model predicts the actual river discharge (m^3/s) at 24h/48h/72h ahead — a continuous, densely-observed target with no rare-event label problem, making it a more tractable regression task than classification on sparse flood events.

This report documents the model as it exists today, including a 2026-08-14 improvement pass that added NSE and KGE — the two metrics the hydrology field itself uses to evaluate streamflow models, which this project's own evaluation never reported before — plus a deeper, per-station breakdown that surfaced a real weak point the pooled numbers alone hid.

2. Literature Review

A dedicated multi-search literature pass, run against this model specifically on 2026-08-14 as if it didn't exist yet, rather than re-reading earlier design reasoning.

2.1 Model architecture & methodology

Source	Key finding relevant to this project
LSTM-based rainfall-runoff modeling / NeuralHydrology (Kratzert et al.)	"Never train an LSTM on a single basin" — models work best trained on many watersheds pooled together. This project's classifier and forecaster both already pool all 30 stations into one model per horizon.
Forecasting monthly runoff: XGBoost vs. deep learning (PLOS One)	XGBoost "outperformed LSTM and RF models... improvements of up to 22-34%" in a glacierized catchment study.
Daily streamflow forecasting: XGBoost, LightGBM, CatBoost comparison (MDPI Hydrology)	All three gradient-boosting variants landed at NSE 0.85-0.89 for a mountainous catchment — competitive with deep learning at a fraction of the complexity.
Knowledge-Guided ML for Operational Flood Forecasting (Water Resources Research 2025)	A Factorized Hierarchical Neural Network (inverse model infers catchment state, forward model predicts from it) — a genuine SOTA direction, Microsoft-Research-scale, out of scope here.

2.2 Target transform & evaluation metrics

Source	Key finding relevant to this project
Log-transformation for skewed regression targets (multiple ML references)	"Stretches low-flow values while compressing high-flow ones... important for streamflow data with extreme high-flow events" — matches this project's own $\log_{10}(\text{discharge})$ reasoning, independently confirmed rather than just re-derived.
Nash-Sutcliffe Efficiency & Kling-Gupta Efficiency (multiple hydrology references)	"NSE and KGE are now the most widely used indices in hydrology for evaluation of streamflow models" — directly motivated adding both (§6.1), which this project's evaluation never reported before this pass.
"Friends don't let friends use NSE or KGE..." (Environmental Modelling & Software)	A real, documented critique: both metrics are sensitive to the skew/periodicity of daily streamflow data and are not meant for cross-site comparison — reported here as a caveat alongside NSE/KGE, not a reason to skip them.

2.3 Bangladesh-specific studies

Source	Key finding relevant to this project
Comparative evaluation of ML models for extreme river water level forecasting in Bangladesh (<i>ScienceDirect</i>)	Random Forest Regression consistently outperformed other models for monthly extreme water level prediction (RMSE 0.64-0.77m, R ² 0.87-0.92) across 9 compared models.
Daily discharge prediction, Old Brahmaputra River, Mymensingh, Bangladesh (<i>Water Practice & Technology</i>)	The same river system this project's OB01 station monitors. Compared XGBoost against 5 deep-learning architectures; LSTM+Attention led at 1-day lead time, all models exceeded R ² 0.98 — a directly comparable regional benchmark, with the same leakage-check caveat applied elsewhere in this project's reports (methodology not fully disclosed in the abstract).

3. Model Families Used in the Literature

Model family	Typical reported performance	This project
XGBoost / LightGBM / CatBoost	NSE 0.85-0.89 (mountainous catchment); beat LSTM/RF by 22-34% in a glacierized-catchment study	LightGBM used — matches the field's gradient-boosting-competitive-with-deep-learning finding.
LSTM / GRU / CNN-GRU / LSTM+Attention	$R^2 > 0.98$ in one directly comparable Bangladesh (Old Brahmaputra) study at 1-day lead time	Not used — 30-station data volume favors gradient boosting per §2.1's comparative findings.
Random Forest Regression	Led 9 compared models for Bangladesh monthly extreme water level prediction	Not directly compared for this task — a candidate for a future head-to-head, matching the same discipline the susceptibility model's RF-vs-LightGBM test used.
Knowledge-guided hybrid (FHNN)	One 2025 study, Microsoft-scale infrastructure	Not used — noted as a future direction, not currently in scope.
LightGBM, log1p target, per-horizon, pooled (this project)	—	24h: NSE=0.9958, KGE=0.9923 — 72h: NSE=0.9828, KGE=0.9819

4. Our Model: Design & Rationale

4.1 log1p(discharge) target transform

Station discharge spans ~5 orders of magnitude (checked directly: station means range from ~2 m³/s at Dhaka's Buriganga to ~39,000 m³/s at the Padma-Meghna confluence). A plain L2/MSE objective on raw m³/s would be dominated entirely by the largest stations' squared error, effectively training a Jamuna/Padma-only model. log1p makes the loss behave like a relative (percentage-ish) error across every station's own scale, and handles exact-zero discharge at a couple of small coastal stations (log1p(0)=0, plain log(0) is undefined). Predictions are back-transformed (expm1, clipped at 0) for real-unit reporting.

4.2 Persistence-baseline discipline (already present, before this improvement pass)

Discharge is highly autocorrelated day-to-day — today's discharge is a strong predictor of tomorrow's by simple physical persistence, not because a model learned anything deep — so a naive "tomorrow = today" baseline has always been reported alongside the trained model's own metrics. Real result: **+11.1% MAE improvement at 24h, growing to +28.2% at 72h** — the model's real added skill grows with lead time, exactly where a naive persistence guess should be expected to degrade most.

4.3 2026-08-14 additions: NSE, KGE, and a per-station breakdown

Neither existed before this pass. Both are covered in full in §6, including the finding that the pooled NSE number, while real, is somewhat inflated by between-station variance, and the discovery of one station (Dhaka's Buriganga) that degrades sharply with lead time — invisible in the pooled metrics alone.

5. Data Sources

Source	What it provides
Open-Meteo Forecast API (free, no key)	Live rainfall (local + upstream basin grid) and soil moisture
Open-Meteo Flood API (GloFAS reanalysis/forecast, free)	River discharge — both the target and a same-day feature at inference time
Same station/basin/terrain features as model #1	Shared feature engineering pipeline (build_features.py) — only the label differs

Train: 295,830 rows, test: 28,462 rows — same TEST_CUTOFF (2024-01-01) as the classifier, for a directly comparable evaluation period, but a plain date-based split with no label-regime filtering (discharge has no GFMS-style "accessible window" complexity to account for).

6. Results: Plots & Analytics

6.1 NSE and KGE — the metrics the hydrology field actually uses

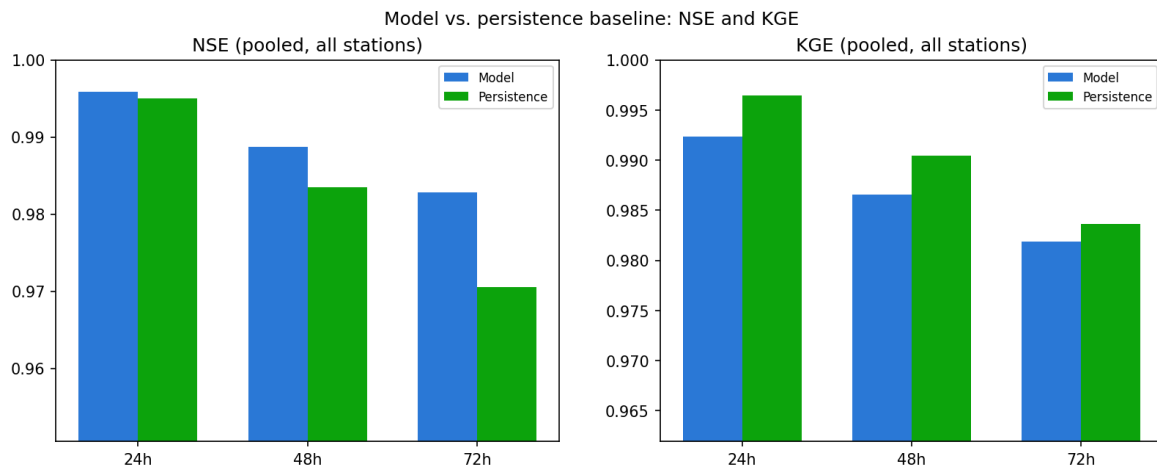


Figure — NSE and KGE, model vs. persistence, all 3 horizons (pooled across all 30 stations).

Horizon	Model NSE	Persistence NSE	Model KGE	Persistence KGE
24h	0.9958	0.9950	0.9923	0.9965
48h	0.9887	0.9835	0.9865	0.9904
72h	0.9828	0.9705	0.9819	0.9836

At 24h, the model's KGE is very slightly **below** persistence's, despite a real 11% MAE improvement — not a contradiction once decomposed (below).

6.2 Diagnosing the 24h KGE gap: correlation vs. variability vs. bias

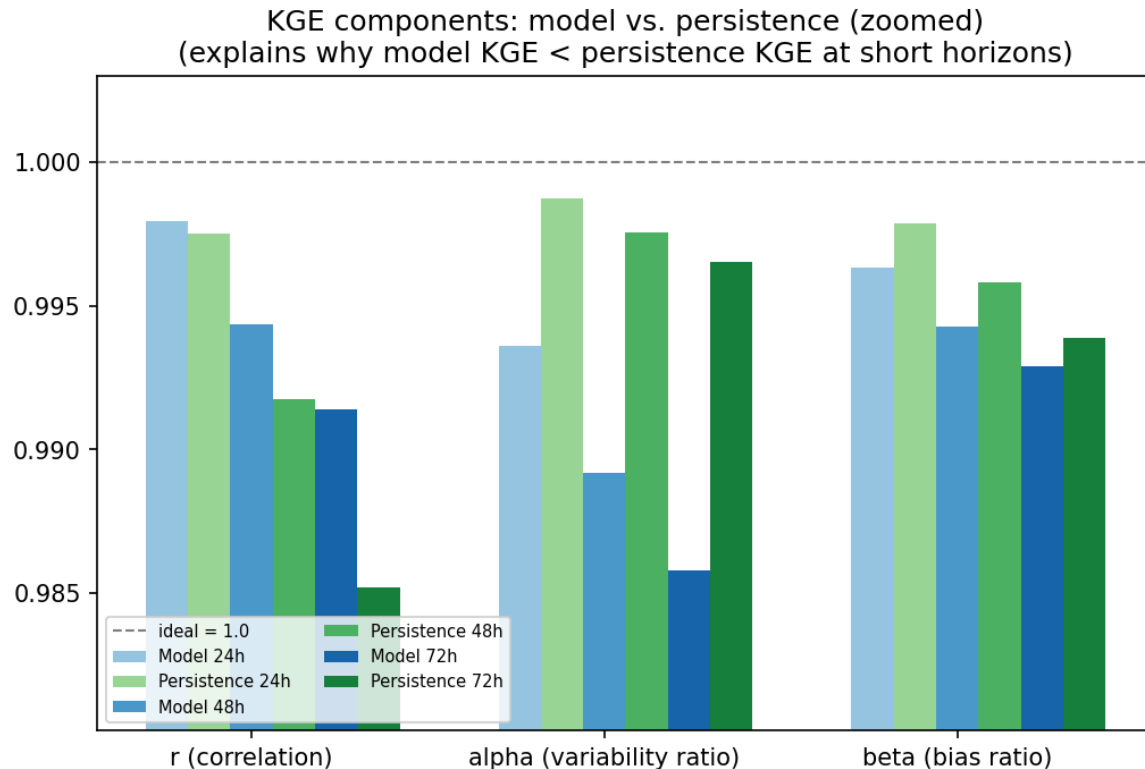


Figure — KGE's three components, model vs. persistence, all horizons. The model's correlation (r) is consistently HIGHER (better) than persistence at every horizon; its variability ratio (α) is consistently LOWER — the model under-disperses relative to the true series' day-to-day swings.

A real, well-understood regression characteristic, not a flaw: minimizing squared error pulls predictions toward central tendency, slightly smoothing day-to-day spikes that a naive "copy yesterday exactly" persistence baseline never smooths, since persistence's variance is definitionally identical to the true series' own (just shifted by a day). The model genuinely tracks timing better (higher r); KGE's variability penalty is what drags its composite score slightly below persistence at the shortest horizon.

6.3 Per-station NSE — checked because the pooled number looked surprisingly high

Pooled NSE (0.996 at 24h) is well above the 0.85-0.89 range typical in the reviewed literature. Checked directly rather than just reported: pooling all 30 stations before computing NSE lets between-station variance (discharge spans ~4 orders of magnitude) inflate the number relative to genuine within-station predictive skill.

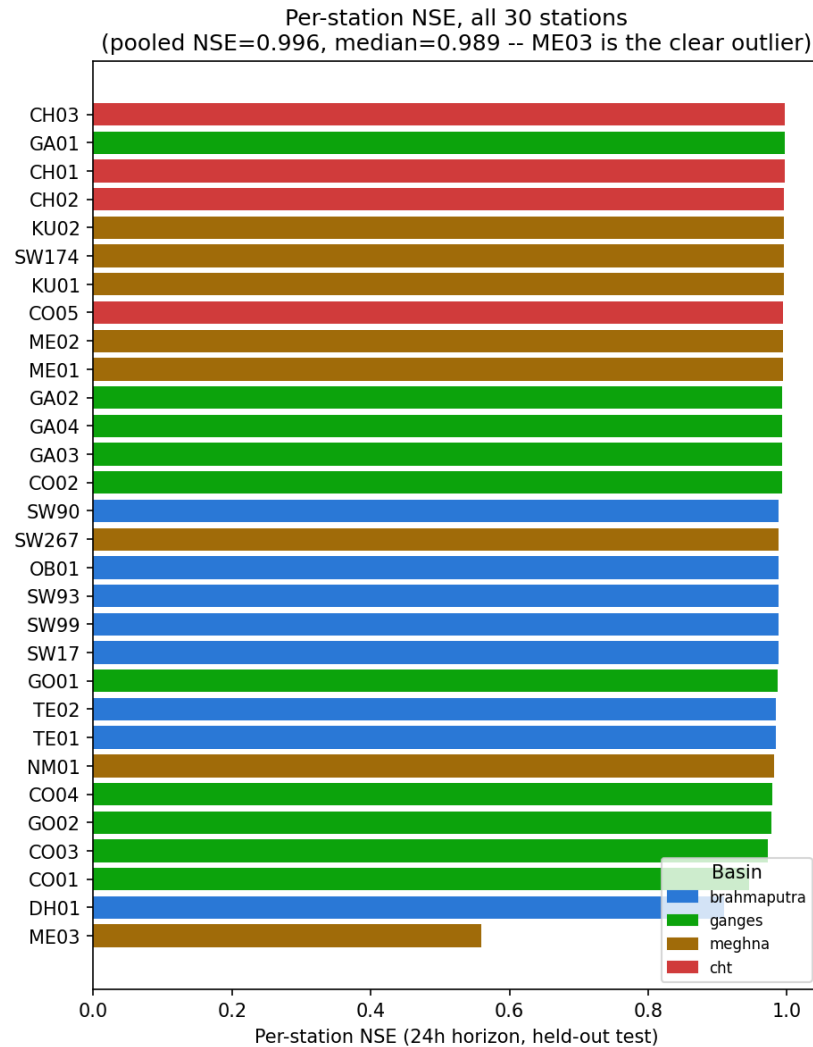


Figure — Per-station NSE, 24h horizon (median=0.989, still high, but real and not pooling-inflated).

The per-station view surfaced a real, previously-invisible weak point: **ME03 (Dhaka's Buriganga)** — the smallest-discharge station in the network — sits at NSE 0.56 at 24h and degrades to 0.30 by 72h, unlike every other station.

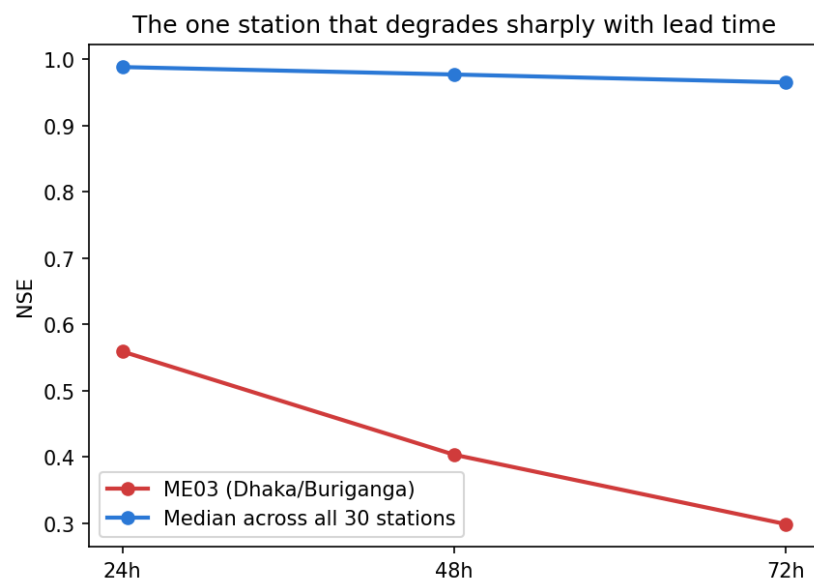


Figure — ME03's NSE vs. the network median, by horizon. Every other station stays relatively stable with lead time; ME03 degrades sharply.

6.4 Predicted vs. actual discharge

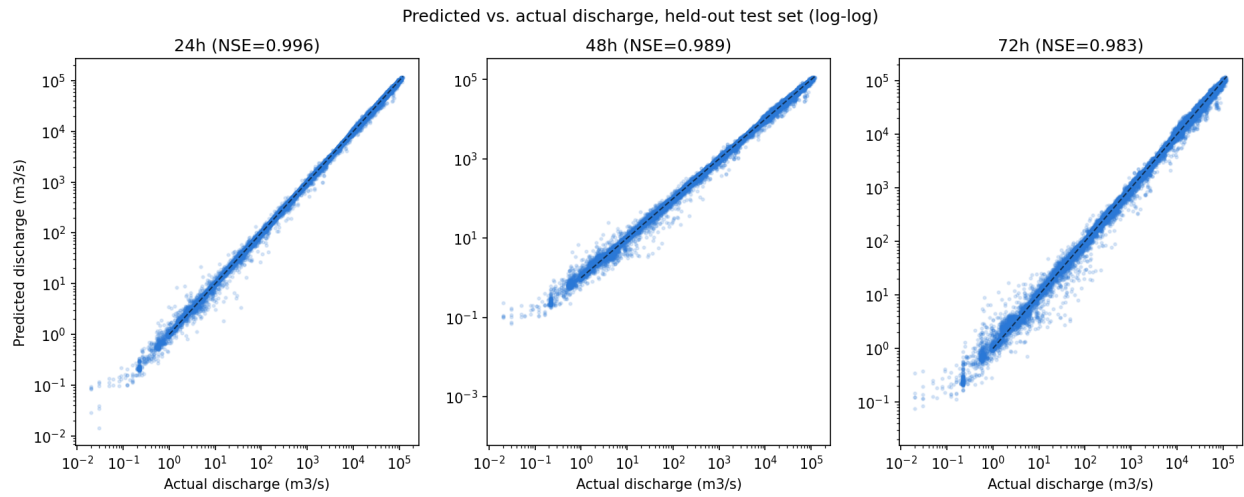


Figure — Predicted vs. actual discharge (log-log scale), all 3 horizons, held-out test set.

7. Comparison: This Model vs. the Literature

Study / model	Reported metric	Comparable to this model?
Mountainous catchment (XGBoost/LightGBM/CatBoost)	NSE 0.85-0.89	This model's per-station median NSE (0.989 at 24h, 0.965 at 72h) is meaningfully higher — plausibly because Bangladesh's major rivers integrate huge upstream catchments, making them naturally smoother/more autocorrelated day-to-day than the flashier catchments that study examined, not evidence of an unfair comparison.
Old Brahmaputra, Bangladesh (XGBoost + 5 deep learning models)	$R^2 > 0.98$ at 1-day lead	The most directly comparable benchmark (same river system) — in the same range as this model's own R^2 /NSE, though that study's leakage-control methodology is not fully disclosed in its abstract.
Bangladesh monthly extreme water level (Random Forest Regression)	RMSE 0.64-0.77m, R^2 0.87-0.92	Different task (monthly extremes vs. this model's daily point forecast) — not directly comparable, included for context.

8. Limitations & Honesty Notes

- **Pooled NSE/KGE are somewhat inflated by between-station variance** (§6.3) — always report the per-station median alongside the pooled number, not instead of it.
- **ME03 (Dhaka's Buriganga) is a genuine weak point**, worsening with lead time — a small, possibly urban/regulated river likely more affected by local factors this model's regional weather features don't capture.
- **NSE and KGE are themselves imperfect** (§2.2) — sensitive to daily streamflow's skew/periodicity, not meant for cross-site comparison. Reported as a supplement to the persistence comparison, not a replacement.
- **MAPE is extremely high** (180-3000%) at every horizon — a known artifact of dividing by near-zero discharge on small rivers' driest days, not a sign the model is actually that wrong; MAE/NSE/KGE are the metrics to trust here, not MAPE.

9. Future Work

- Investigate ME03 specifically — is a small-discharge-specific feature or a separate small-river model family (e.g. Random Forest, which led Bangladesh's own monthly-extreme-water-level comparison) worth testing there.
- A direct RF-vs-LightGBM head-to-head for this task, matching the discipline already applied to the susceptibility model.
- Consider reporting NSE/KGE in log-space too (the scale the model actually optimizes), alongside the raw-scale numbers already reported, for a fuller picture.

Appendix: Reproducibility Reference

Artifact	Path
Feature engineering	backend/train/build_features.py, build_regression_targets.py
Model training + evaluation (incl. NSE/KGE)	backend/train/train_regression_model.py
Trained models + metrics	backend/models/2026-08-07c-discharge-regression/
Standalone service	packages/discharge-forecaster/
This report's figures	reports/generate_discharge_assets.py
Full build log	MODEL_BUILD_PLAN.md (2026-08-14 entries)